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Title: Security and Privacy in E-Health Systems: A Review of AI and Machine Learning Techniques

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Summary:

The paper introduces e-health systems as comprehensive frameworks that integrate electronic and information technologies to deliver healthcare services, manage health information, and enhance patient outcomes. It traces the evolution of e-health from cumbersome paper-based records to advanced Electronic Health Records (EHRs), highlighting the shift driven by information and communication technologies (ICT) in the late 20th century. Key benefits include improved access through telemedicine, personalized medicine via data analysis, cost efficiency by reducing hospital visits, patient empowerment through portals, secure data management, and public health advancements like disease surveillance. Major components discussed encompass EHRs and EMRs, telemedicine platforms, Health Information Exchange (HIE) networks, Clinical Decision Support Systems (CDSS), wearables, patient portals, health analytics tools, mHealth applications, and regulatory frameworks. The introduction emphasizes the growing cybersecurity spending in healthcare, projected to exceed 27 billion USD by 2026, underscoring the need for robust protections amid rising threats.

Key topics cover evolving threats in healthcare data security, such as a 239% increase in hacking incidents and 278% rise in ransomware attacks from 2018 to 2023, with over 133 million records exposed in 2023 alone, including major breaches like Anthem Inc. (2015) and Change Healthcare (2024). Vulnerabilities in e-health infrastructure are analyzed, focusing on cloud computing (e.g., data breaches, unauthorized access via phishing, keylogging, man-in-the-middle attacks, brute force, and credential stuffing; compliance issues with HIPAA), EHRs (cyber attacks, outdated software, phishing), Internet of Medical Things (IoMT), and mHealth applications. The paper explores AI and machine learning (ML) techniques for threat detection and prevention, including supervised and unsupervised learning models for anomaly detection, automated countermeasures, and adaptive algorithms. Privacy-preserving methods like federated learning, differential privacy, and secure multiparty computation are highlighted, alongside secure model deployment, blockchain for data interoperability, ethical considerations, regulatory compliance, and real-world applications of AI/ML in securing e-health systems. Emerging trends include large language models (LLMs) for healthcare transformation, their use cases, challenges, and quantum-resistant encryption.

The methods involve a comprehensive review of literature and trends, synthesizing data on breaches, vulnerabilities, and AI/ML applications without original empirical experiments. Findings reveal that AI enhances security through continuous monitoring, pattern recognition, real-time responses, predictive analytics, and scalability, significantly improving threat detection efficiency while addressing privacy

concerns. Conclusions stress future research opportunities, such as adaptive threat detection systems, privacy-preserving AI, automated compliance, blockchain-secured interoperability, and ethical AI development. By prioritizing innovative solutions, collaboration, and ethical practices, the e-health sector can foster resilient infrastructures that protect sensitive data, build patient trust, and ultimately improve healthcare delivery and outcomes.